

Rishindra Davuluri

☎ 346-410-8427 | ✉ rishdavulu@utexas.edu | 🔗 linkedin.com/in/rishindradavuluri | 🐙 github.com/rishindrum

EDUCATION

The University of Texas at Austin *B.S., Computer Science Honors* Austin, TX
Texas CSB Honors Program *B.B.A., Canfield Business Honors* Aug. 2023 - May 2027

Relevant Coursework: Machine Learning II (H), Principles of ML I, Algorithms (H), Java Data Structures (H), C++ Operating Systems (H), Computer Architecture w/ C (H), Object-Oriented Programming, SQL MIS (H), Data Science in R (H), Decision Science (H), Discrete Math (H), Linear Algebra, Multi-variable Calculus, Statistics & Probability **(H): Honors**

Work Eligibility: US Citizen

EXPERIENCE

UT Austin: Learning Directed Operating System - *Undergraduate ML Researcher*; Austin, TX Oct 2025 - Present

- Designing low-latency **Model Context Protocol** in **C** to synchronize decision-making across learned OS components
- Benchmarking protocol on **CloudLab** bare-metal instances using conflicting learned agents (e.g., Prefetcher vs. Allocator)

Moody College of Communication - *Undergraduate NLP Researcher*; Austin, TX Nov 2025 - Feb 2026

- Categorized narratives via anchor-based semantic classification using **sentence embeddings** and **cosine similarity**
- Engineered **BERTopic** and **RegEx** feature-engineering pipeline to analyze demographic disparities in financial complaints

Capital One - *Software Engineer Intern*; McLean, VA Jun 2025 - Aug 2025

- Architected cache library and **Snowflake** dataset using **Python** and **AWS Lambda** for omni-channel experience
- Integrated cache into **Vue.js** and **Vuex** webpage, improving workflow completion by **40%** for **80K** monthly fraud concerns
- Built educational phishing simulation platform using **Next.js** and **Twilio** to send custom emails and track user clicks

UT Austin: Dynamic Belief Games - *Undergraduate ML Researcher*; Austin, TX Sep 2024 - Jan 2025

- Trained and tested **GCNNs** and **PyTorch SB3** reinforcement learning models to optimize network communication
- Leveraged **NumPy**, **Pandas**, and **Matplotlib** to present data analysis to **U.S. Army Futures Command's UTDD**

Fidelity Investments - *Software Engineer Intern*; Westlake, TX Jun 2024 - Aug 2024

- Enhanced batch responsible for **100M+** updates to central DB with **AWS Q Developer, SQL, & MyBatis**
- Accelerated processing by **20%** by migrating 30+ files from **C++** to **Java** for multi-threading and cloud integration

PROJECTS

Ace: Proactive AI Study Agent | *Go, Python, Gemini API, Angular, gRPC, Google Cloud, Neo4j, BigQuery*

- Architected **Angular** UI and **Go/Python** microservices mesh on **Cloud Run**, using **gRPC** for low-latency syllabus analysis
- Constructed **RAG** pipeline with **BigQuery** and **Neo4j** to automate personalized quizzes based on student "struggle areas"
- Automated study weekly session scheduling via **Google Calendar API** and persistent **GCS**-based vector index management

Operating System & Graphics | *C++, Python, Assembly, QEMU, Bash, Linux, Multi-threading*

- Developed core **C++** and **Assembly** kernel: interrupt handling with $O(1)$ helper threads; fork/wait system calls; semaphores
- Engineered virtual memory, **Ext2** file system with third-level indirection, and VBE graphics subsystem with **Python** frame conversion

RestoreAI | *Python, Pandas, NumPy, Random Forest, Gemini API, React, JavaScript, RESTful APIs*

- Selected **Random Forest** model in Python using **Pandas** and **NumPy** after cross-validation and hyperparameter tuning
- Streamlined React UI to ML model integration using **RESTful API** architecture and **Gemini API** function calling

LEADERSHIP EXPERIENCE AND ACTIVITIES

Texas Product Engineering Organization - *Engineering and Product Fellow*; Austin, TX Sep 2024 - Present

- Developing **Python** webcrawlers, **PHP** plugins, and websites with **React**, **REST**, and **Vercel** for nonprofits
- Built **Locus** with **TS**, **Node.js**, and **Firebase** (**Realtime DB/Firestore**) for **sub-100ms** location and status updates

Texas Convergent - *Forge Tech Member, Meals on Wheels and Cura Project Intern*; Austin, TX Aug 2023 - Sep 2024

- Engineered **RAG** pipeline integrating **GPT-4**, **FastAPI**, and **Supabase** for personalized patient care plans
- Deployed **React** app and **Flask** backend for **50+** drivers, integrating Salesforce (**SOQL**) for maps and task updates

ADDITIONAL INFORMATION

Technical Languages: Python, C++, TypeScript, Go, Java, C, JavaScript, C#, R, SQL, SOQL, CQL

Cloud & DB: GCP (GCS, GC Run, BigQuery, Firebase), AWS (Lambda, S3, EC2), Terraform, Snowflake, Neo4j, Apache Cassandra

Tools: Gemini, PyTorch, NumPy, Pandas, sklearn, Matplotlib, Angular, React, FastAPI, Salesforce, Linux, Docker, GitHub Actions

Certifications: Eagle Scout | Azure AI Fundamentals | NLP & Text Mining (NLTK, BERTopic) by Simplilearn

Service & Interests: Texas Infinites (Spirit and Service); Singing & Guitar; Camping; Portrait Drawing; Volleyball & Soccer

Languages: Native in English and Telugu; Conversant in Spanish